Monday Warm Up: Unit 3, Magnetic Forces and Fields

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1 Memory Bank

- $\vec{F} = q\vec{v} \times \vec{B}$ ... The Lorentz force, one version.
- $\vec{F} = i\vec{L} \times \vec{B}$ ... The Lorentz force, another version.
- $\vec{\tau} = \vec{r} \times \vec{F}$ ... Definition of torque.
- $\vec{\tau} = \vec{\mu} \times \vec{B}$ ... Torque on a current carrying loop in a uniform B-field.
- $\vec{\mu} = NI\vec{A}$ ... Definition of the magnetic moment.
- $\vec{B} = \mu_0 I/(2\pi r)\hat{\phi}$ ... B-field created by a straight segment of wire, Ampère's Law.

2 Torque, revisited

1. Suppose we apply a force $\vec{F} = 10\hat{j}$ N (upward) on the end of a 10 cm wrench it as *loosens* a bolt at the origin. The bolt is to the left of the wrench. What is the magnitude *and direction* of the torque?

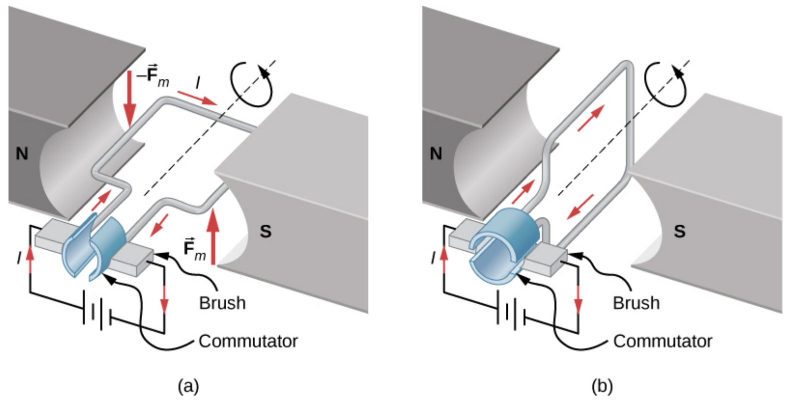

Figure 1: An illustration of how a power generator works. This version uses DC current and a commutator.

2. Consider Fig. 1, in which a DC electric motor is depicted inside a 0.05 T B-field. Suppose the area of the loop is 10^{-2} m^2, the voltage in the circuit is 24 V, and the circuit resistance is 50 Ω. Also assume that there is just one loop of wire in the rotor. What is the *maximum torque* the system could achieve?

3 The Electric Motor

1. Let us derive the maximum torque for an electric motor like the one shown in Fig. 1. The magnetic moment of a loop of current is $\vec{\mu} = NI\vec{A}$, where N is the number of turns, I is the current, and $\vec{A}$ is the area vector. (a) Based on Fig. 1 (left), show that the torque $\vec{\tau}$ on the current loop is $\vec{\mu} \times \vec{B}$. (b) Given what you know about the cross-product, what is the maximum magnitude of $\vec{\tau}$?

3. What would the maximum torque be if there were $N = 100$ turns of wire?

4 Ampère's Law

1. What is the magnitude of the B-field created by a 1 A current within a wire, 1 cm away from the wire? How does this compare to the B-field of the Earth?